BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-II B. Sc. Engineering Examinations 2023-2024

Sub: **CSE 213** (Software Engineering)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer to **Q. No. 1** is mandatory.

Answer any **TWO** questions from the rest.

1. (a) A travel company plans to offer customers a number of travel packages. Each package has its own set of activities, pricing, and a brief description. Currently, the system supports two package types: **(15)**

Adventure Package: Designed for thrill-seekers, this package includes physically engaging outdoor activities such as mountain hiking, river rafting, etc.

Regular Package: Designed for regular leisure travelers, this package includes city sightseeing, museum visits, etc.

Each package independently manages its own details. Your task is to implement the tour package management system with an appropriate design pattern so that the client code does not directly construct a concrete class, but the appropriate package object is instantiated through a centralized creation component. Moreover, the design must support new package types without modifying any existing code.

(b) A software company developed an online citizen information platform for a government body. After the first release, the corresponding team hired a security engineer to identify and resolve security vulnerabilities within the system. **(10)**

During the assessment, the engineer discovered several critical flaws that posed high risks of exposing sensitive citizen data. Immediately after the discovery, he disclosed the vulnerabilities to media outlets anonymously. He justified his actions by claiming it was his moral duty to protect citizens who are already using this from potential harm. The resulting media coverage triggered a public scandal that severely damaged the firm's reputation and disrupted the government project. Investigate whether there is any ethical breaches present in the above scenario with proper justification.

(c) A startup client approaches a software development team with the idea of a mobile application called "Personalized Mental Health Companion". The application is expected to track users' mental states, provide AI-based mental exercises, and include a virtual companion feature. The client has only provided high-level requirements and left detailed design and implementation decisions to the development team. To manage development, the team chooses an incremental process model. They plan to build the system in small functional parts so that they can get client feedback after each increment. Investigate the suitability of the incremental model for this scenario, with justification, and propose alternative process models if you find another more appropriate. **(10)**

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2. (a) FinCorp is a financial consultancy firm that originally operated with one branch in several states across the United States. Every month, the headquarters received state-wise expenditure reports of operational expenses. Recently, FinCorp expanded significantly. It now operates multiple offices across different cities within each state and has also expanded internationally with offices in several European countries. As a result, reporting requirements have become more complex. Different stakeholders require expenditure summaries at multiple levels: **(20)**

- A State Director needs a consolidated report for all cities within a state
- A Country Head needs a report aggregating all states within a country
- The Global CFO needs a complete worldwide report

Each office (city, state, country) maintains its own expenditure data. Higher-level reports must aggregate data from all subordinate units, perform analysis on the aggregated data, and attach the results to the report. The nature of this analysis and processing may vary across different organizational levels.

Besides, some additional sections might be provided with the original report, such as, tax calculation details, investment summaries, and government compliance reports. These additional sections may or may not be included, and a single report may include one or more of them based on instructions, without changing the core reporting structure.

As a member of the development team, you must redesign the reporting system so that it supports (i) the hierarchical structure while keeping the reporting interface consistent for all levels and (ii) the flexible inclusion of additional report sections based on requirements.

Identify the appropriate design pattern(s) to satisfy these requirements, and provide an implementation with the minimum required functionality.

(b) You and your friend are co-founders of an online education platform. After initial success, the platform attracted several investors. With this funding, you planned to introduce a major AI-based feature that users have been demanding for a long time, especially as some competitors are already moving in the same direction. The feature is expected to significantly boost user engagement and growth. **(15)**

However, due to political and economic unrest, you anticipate a potential financial crisis. Therefore, you want to cancel or postpone the feature to conserve resources. On the other hand, your partner believes that launching the feature could attract more users and generate revenue to offset crisis.

Identify the risks associated with both strategies. Categorize these risks based on their nature and analyze heir potential impact on the platform's future.

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3. (a) Suppose, you are managing a software project named "SmartHome" that consists of three major tasks. The planned progress and current status of all tasks are shown in Figure 1. At the end of Week 3, you are analyzing the schedule and budget of the project using Earned Value Management. You are impressed that the second task has already been completed by Week 3, i.e., ahead of the planned schedule. However, the status of the third task remains unknown as it was assigned to a freelancer who has not provided any clear updates yet and this made you anxious. **(20)**

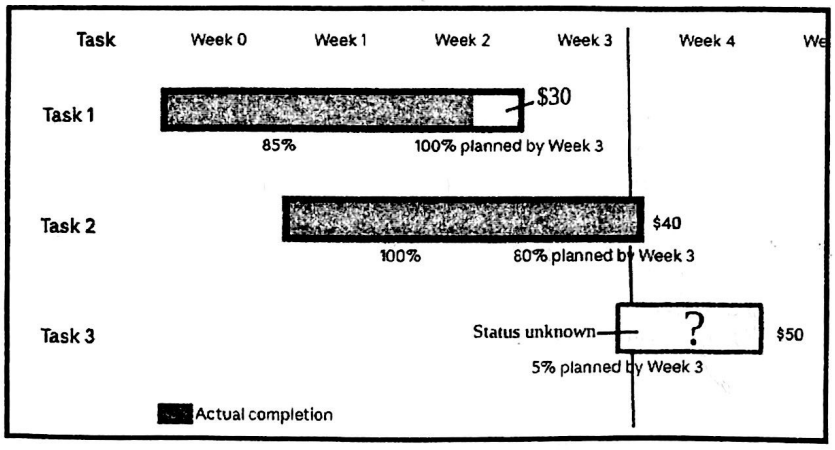


Figure 1: Gantt Chart of SmartHome

To calm you down, your project officer claims: "Whatever the progress of the third task is, the project is ahead of schedule."

However, you could not relax as after a bit of investigation about actual cost, you find out that: "No matter what the progress of the third task is, the project will exceed the total budget."

(i) Verify whether the project officer's claim about the schedule is correct.

(ii) Based on your findings, find the lower bound of actual cost (AC).

(b) An online library platform provides access to a large collection of digital resources, including books, journals, research papers, and multimedia content. Users can search the catalog, view details, read content, and interact through features such as bookmarking and reviews. As the platform grows in popularity, it experiences a high volume of read requests (e.g., searches and content access), while write operations remain relatively low. At the same time, the continuously expanding database is creating challenges in scalability and performance. **(15)**

To address these issues, the company is considering implementation of server-side caching and database sharding. Additionally, an investor has suggested that, given the platform's expected global reach, adopting a Content Delivery Network (CDN) in the near future would be beneficial. Discuss whether and how these approaches would effectively address the platform's challenges and improve performance.

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4. (a) Identify 5 (five) bad smells present in the following code snippet and perform refactoring for each of them. **(20)**

```
class Student {
  String name, id;
  int type;          // 1 = UG, 2 = PG
  int year;
  String street, post office, district;
  float[] yearWiseGpa;
  Bool isFulltime; // For PG

  // constructors ...

  float calculateFee() {
    if (type == 1) {              // UG
      return 50000;
    } else if (type == 2) {       // PG
      float sum = 0;
      for (float gpa : yearWiseGpa) sum += gpa;
      float cgpa = sum / yearWiseGpa.length;
      if (cgpa > 3.5) return 40000;
      else        return 48000;
    }
    return 0;
  }

  void printDetails() {
    // print common details
    System.out.println("Name:" + name);
    System.out.println("Student ID:" + Id);
    System.out.println("Address: " + street + ", " + postOffice + ", " + district);
    ...
    ...
    if (type == 1)
      // Prints UG student-specific details
    else {
      // Prints PG student-specific details
    }
  }
}

class Scholarship {
  String name;
  int   amount;
  float  CgpaThreshold;

  // constructors ...

  boolean isRegular(Student student) {
    // Returns true if the student has no gaps in yearly GPA records
    return student.yearWiseGpa.length == student.year;
  }
```

Contd P/5

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Contd.... for Q. No. 4(a)

```
void checkEligibility(Student student) {
if (stuent.type == 1)
{
  float sum = 0;
  for (float gpa : yearWiseGpa) sum += gpa;
  float cgpa = sum / yearWiseGpa.length;

  if (isRegular(student)) {
    if (student.year >= 2) {
      if (cgpa >= CgpaThreshold) {
        System.out.println(student.name + " is awarded scholarship: " + amount);
      } else {
        System.out.println(student.name + " is not eligible..");
      }
    } else {
      System.out.println(student.name + " is not eligible.");
    }
  } else {
    System.out.println(student.name + " is not eligible.");
  }
 }
}
```

(b) Define code smell, tech-debt and code refactoring. Differentiate between proactive and reactive risk management strategies with examples. **(15)**

SECTION – B

There are **FOUR** questions in this section. Answer to **Q. No. 5** is mandatory.
Answer any **TWO** questions from the rest.

5. (a) A university course-registration system decides whether a student may register for a lab section based on the following three independent inputs: **(15)**

Completed Credits	**CGPA**	**Status**
C1: less than 60	**G1:** below 2.50	**S1:** not approved by advisor
C2: 60 to 99	**G2:** 2.50 to 3.49	**S2:** approved by advisor
C3: more than 99	**G3:** 3.50 or above	**S3:** invalid or missing status
C4: invalid credit input	**G4:** invalid CGPA input	

Using Equivalence Class Testing (ECT), perform the following:

(i) Construct a minimal weak ECT test suite for this system.
(ii) Compute the total number of test cases required for strong ECT.
(iii) Illustrate one defect scenario that may remain undiscovered if only the weak ECT suite is used.

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Contd.... for Q. No. 5

(b) A Smart Home Hub coordinates communication among devices such as motion sensors, lights, alarms, and other smart devices. **(10)**

(i) Suppose a motion sensor event may trigger one or more coordinated actions, such as turning on a light, activating an alarm, or notifying the hub. Illustrate how the Mediator pattern is preferable here than allowing the Motion Sensor to communicate with the other device objects directly.

(ii) Suppose they system must support recording, queueing, and later replaying of sensor-triggered actions such as turning on a light, activating an alarm, or sending a notification. Select a suitable behavioral design pattern for this requirement and show how it can be applied.

(c) Suppose you join a startup and find that it has already developed a software product without preparing any formal written specification. The company now wants you to prepare a proper specification so that the system's intended behavior is clearly documented and can serve as an agreed reference for future review and improvement. Explain how you would proceed. **(10)**

6. (a) A Java method isLeapYear(int year) is already implemented. It returns true for leap years, false for non-leap years, and throws an IllegalArgumentException if year ≤ 0. Write a complete JUnit test class for this method. Your tests should cover all logically distinct input classes relevant to leap-year determination, including invalid input. Use meaningful test method names and appropriate assertions. **(15)**

(b) During online registration night, the university portal becomes unavailable. One engineer argues that the failure is due to ordinary overload caused by many legitimate student requests at the same time. Another argues that it may be a Denial-of-Service (DoS) attack. **(10)**

(i) Explain how these two possibilities can be distinguished in practice.

(ii) Briefly discuss two defenses that could help mitigate the problem.

(c) A smart temperature system has the following three modules. **(10)**

- Display Panel: top-level module that shows temperature and warning messages to the user
- Temperature Controller: middle-level module that receives sensor data, processes it, and decides what should be displayed
- Sensor: low-level module that reads the actual temperature from the environment

The Display Panel and Sensor are ready first, but the Temperature Controller is still under development. How would you conduct integration testing in this situation?

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7. (a) You are asked to prepare a performance testing plan for a course-registration system. The system faces the following operational risks: **(15)**

- a sudden burst to 10,000 students when registration opens
- possible collapse when the load exceeds the system's capacity
- gradual slowdown after 8 hours of continuous operation

For each of the above situations:

(i) Identify the most appropriate performance testing type and briefly state the goal of that test.

(ii) Propose a simple JMeter testing configuration by specifying Threads, Loop Count, Ramp-Up, and Duration.

(b) A smart irrigation system activates a water pump based on an integer moisture sensor reading M. The specification is as follows: **(10)**

- Pump is OFF if $M < 30$
- Pump runs at LOW power if $30 \leq M \leq 70$
- Pump runs at HIGH power if $M > 70$

Derive the complete set of test cases for the LOW-power partition using Boundary Value Analysis.

(c) Consider the following login program that checks a password and sets an authentication flag accordingly. **(10)**

```
#include <stdio.h>
#include <string.h>

struct LoginData {
    char password[8];
    int isAuthenticated;
};

void check_login(const char *input) {
    struct LoginData data;
    data.isAuthenticated = 0;

    strcpy(data.password, input);

    if (strcmp(data.password, "pass") == 0) {
        data.isAuthenticated = 1;
    }
    if (data.isAuthenticated) {
        printf("Login Successful. Welcome!!!\n");
    } else {
        printf("Wrong Password! Login denied\n");
    }
}
int main() {
    char input[8];
    scanf("%s", input);
    check_login(input);
    return 0;
}
```

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Contd.... for Q. No. 7(c)

(i) Identify the security vulnerabilities present in the program.

(ii) Give a practical attack input idea and explain how it may exploit the program.

(iii) Suggest safer alternatives to prevent the problem.

8. (a) Consider the following design for a file-parsing framework: **(15)**

```
abstract class DataParser {
    public final void parse(String fileName) {
        open(fileName);
        extract();
        close();
    }
    protected void open(String fileName) {
        System.out.println("Opening file: " + fileName);
    }
    protected abstract void extract();
    protected void close() {
        System.out.println("Closing file");
    }
}
class CsvParser extends DataParser {
    protected void extract() {
        System.out.println("Extracting data from CSV");
    }
}
class JsonParser extends DataParser {
  protected void extract() {
        System.out.println("Extracting data from JSON");
    }
}
```

The design follows the Template pattern, where the overall parsing workflow is fixed as: open( ) → extract( ) → close( ). Currently, different file formats, such as CSV and JSON, are handled by creating separate subclasses that override the extract( ) method. This approach has a limitation: every new file format requires a new subclass, even though the overall parsing workflow remains the same. Redesign the solution and write the relevant Java code to demonstrate how the Strategy pattern can be used within the Template pattern so that the extraction logic becomes interchangeable at runtime.

(b) A new agency needs a system where multiple displays (Website, Mobile App, and Office Billbaord) are updated instantly whenever a new "Breaking News" headline is published. What is the most appropriate Behavioral Design Pattern for this scenario? Justify your answer. **(10)**

(c) Compare State and Strategy patterns. Why are they structurally similar but conceptually different? **(10)**

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